

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Third Semester B.E. (CSE/ISE) (Credit System) Degree Examinations**

April 2023

21CS303 / 21IS303 - COMPUTER ORGANIZATION AND ARCHITECTURE

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III

Unit – I

		Marks	BT*	CO*	PO*
1	a) Explain the basic functional units of a computer with neat diagram.	5	L*2	1	1
	b) Explain the need of Autoincrement addressing mode with an example.	5	L2	1	1
	c) With the help of neat diagram explain Sequential multiplication process by multiplying 13*11.	10	L3	2	1
2.	a) List and explain the role of Processor registers during memory interaction.	5	L1	1	1
	b) What are Condition codes? List different condition code flags with their meaning.	5	L1	1	1
	c) With the help of neat diagram and algorithm perform 13/3 using restoring division algorithm.	10	L3	2	1
3.	a) Multiply 13 and -6(Multiplier) using Booth's bit pair recoding algorithm.	5	L3	2	1
	b) Represent 52.21875 ₍₁₀₎ in IEEE single precision floating-point number system.	5	L3	2	1
	c) Define addressing mode. Explain any 4 addressing mode with proper syntax.	10	L1	1	1

Unit – II

4.	a) Define ROM. Compare different types of ROMs.	5	L1	4	1
	b) Discuss Associative Mapping technique.	5	L1	4	1
	c) Draw the block diagram of single bus architecture. Write the control sequence to add the contents of register R ₀ to R ₁ and store the result in R ₁ .	10	L3	3	1
5.	a) List different cache replacement algorithms. Explain any two in detail.	5	L2	4	1
	b) With the help of control sequences explain the process of execution of a branch instruction in single bus architecture.	5	L2	3	1
	c) Draw the block diagram of 3-bus architecture. Write the control sequence for Add R ₁ , R ₂ , R ₃ .	10	L3	3	1
6.	a) Discuss Hardwired Control unit with a block diagram.	5	L2	3	1
	b) Explain Address translation process in Virtual memory.	5	L2	4	1
	c) Explain with neat diagram internal organization of 2M*8 dynamic memory chip.	10	L2	4	1

Unit – III

7.	a) What are interrupts? Explain how nested interrupts and simultaneous interrupts are handled.	10	L2	5	1
	b) What is Direct Memory Access? Explain the role of DMA controller during data transfer.	5	L2	5	1
	c) Explain the concept of Synchronous bus in data transfer operation.	5	L2	5	1
8.	a) Explain the concept of Centralized arbitration and distributed arbitration techniques with neat diagram.	10	L2	5	1
	b) What are Exceptions? List and explain different types of Exceptions.	5	L1	5	1
	c) Explain the following: i) Universal Serial Bus (USB) ii) Peripheral Component Interconnect (PCI) bus	5	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
